Rate Theory: Cluster Dynamics, Grouping Methods, and Best Practices

101

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Contents

Abstract

Cluster dynamics is a mean field method which is used to simulate the evolution of defect clusters in materials under aging and irradiation conditions. Its popularity is due to the simplicity of the underlying "cluster gas" model, which is able to simulate accurately and efficiently dilute cases, provided it is carefully parametrized. In this chapter cluster dynamics formalism is reviewed. Parametrization of cluster dynamics models and numerical schemes to efficiently solve cluster dynamics equations are discussed.

1 Introduction

Microstructure evolution under irradiation is driven by phenomena occurring over very different time and length scales. For example, displacement cascades are produced in a few picoseconds and span over a few tens of nanometers, whereas subsequent defect diffusion over micrometers requires a few seconds or more.

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Quantitative simulation of the impact of radiation damage over experimental time scales requires the modeling of all these processes. Although no single tool is capable of performing such a task, cluster dynamics has emerged as a versatile tool to explore various consequences of irradiation. In particular, when it is informed with atomistic calculations, it has proven capable of simulating precipitation in dilute alloys (Clouet et al. 2005; Jourdan et al. 2010), void and loop nucleation, growth, and coarsening (Ortiz et al. 2007; Xu et al. 2012b; Moll et al. 2013), without suffering the limitations of fully atomistic calculations.

A distinctive feature of cluster dynamics is its mean field character, which solves the length scale problem. Microstructure evolution is obtained by solving a set of ordinary differential equations (ODEs) on cluster concentrations. However, for some phenomena which are highly localized in space, such as displacement cascades, mean field formalism can be a drawback, and special care must be taken. Concerning the time scale problem, high-frequency events, which limit the efficiency of stochastic methods such as object kinetic Monte Carlo (OKMC), may also hinder the solving of cluster dynamics equations. More precisely, the large scatter in reaction rates leads to very stiff equations, for which dedicated methods must be used. In addition, to simulate large clusters, it can be essential to reduce the number of equations by approximating the initial ODE set. Grouping methods (Golubov et al. 2001a) and Fokker-Planck-based methods (Ghoniem and Sharafat 1980) are useful ways to achieve such reduction.

In this article we first present the cluster dynamics formalism. Then, parametrization of cluster dynamics models is discussed. In the last section, technical aspects for the efficient solving of cluster dynamics equations are presented.

2 Cluster Dynamics Formalism

Cluster dynamics is a mean field approach which considers the microstructure as a collection of clusters. These clusters can diffuse in the matrix and absorb and emit other clusters (Martin 2006; Clouet 2009). For this reason, this model is sometimes called a “cluster gas model.” As such, cluster dynamics is essentially appropriate for dilute cases, where clusters can be distinguished unambiguously. Such cases include the precipitation of secondary phases in dilute alloys and the formation of point defect clusters. It bears much resemblance to OKMC and mostly requires the same input parameters.

Let us consider, for example, the formation of vacancy and self-interstitial clusters under irradiation in a pure metal. Assuming that only one type of cluster exists for each self-defect, point defect clusters can be conveniently identified by a single index n. We use $n < 0$ for a vacancy cluster containing $|n|$ vacancies and $n > 0$ for an interstitial cluster containing n self-interstitials. The evolution equation, for the concentration C_n of mobile clusters in class (n), reads (Hardouin-Duparc et al. 2002; Xu et al. 2012a; Jourdan et al. 2014):

$$\frac{\mathrm{d}C_n}{\mathrm{d}t} = G_n + \sum_{\substack{m\in\mathscr{M}\\ n-m\neq 0}} J_{n-m,n} - \sum_{m\in\mathscr{M}} J_{n,n+m} - \sum_{m\in\mathbb{Z}\backslash\{0\}} J_{m,m+n}$$
$$- \sum_j k_{j,n}^2 D_n \left(C_n - C_{n,j}^{\mathrm{e}}\right). \tag{1}$$

In this equation, G_n and D_n are the production rate and the diffusion coefficient of cluster (n), respectively, and $\mathscr{M}$ is the set of mobile species. The factor $k_{j,n}^2$ represents the sink strength of sink type j which is not explicitly treated with cluster dynamics variables, in analogy with classical rate theory approach (Nichols 1978). In the present case, network dislocations and grain boundaries, for example, are taken into account with sink strengths, whereas vacancy and interstitial clusters are explicitly treated with the dedicated equations on C_n. Equilibrium concentration of (n) near sink j is noted $C_{n,j}^{\mathrm{e}}$.

The flux $J_{n,n+m}$ is a net flux between classes (n) and $(n+m)$, due to the mobility of (m) only. It is defined by:

$$J_{n,n+m} = \beta_{n,m} C_n C_m - \alpha_{n+m,m} C_{n+m}, \tag{2}$$

where $\beta_{n,m}$ and $\alpha_{n+m,m}$ are the absorption and dissociation rates, respectively. The effect of mobility of (n) is taken into account with the term $J_{m,m+n}$ in Eq. (1). Absorption rates are closely related to the sink strengths used in rate theory (Nichols 1978). They take different forms, depending on the cluster type, the presence of long-range interactions (elastic, but also electrostatic interactions for materials with charged defects Kotomin and Kuzovkov 1992), and the presence of an energy barrier for the association of clusters (Waite 1958). For example, if (n) is a cavity in an infinitely dilute system and if elastic interactions between (n) and (m) are neglected, we obtain, for a three-dimensional diffusion of (m) (Waite 1957):

$$\beta_{n,m} = 4\pi(r_n + r_m)D_m, \tag{3}$$

which corresponds to the classical sink strength $k_{n,m}^2 = 4\pi(r_n + r_m)C_n$. In Eq. (3) r_n is the radius of cluster (n). The emission rate of (m) by $(n + m)$ is obtained by imposing that $J_{n,n+m}$ is zero at equilibrium, so

$$\alpha_{n+m,m} = \frac{\beta_{n,m}}{V_{\mathrm{at}}} \exp\left(-\frac{F_{n+m,m}^{\mathrm{b}}}{k_{\mathrm{B}}T}\right), \tag{4}$$

where $F_{n+m,m}^{\mathrm{b}} = F_n^{\mathrm{f}} + F_m^{\mathrm{f}} - F_{n+m}^{\mathrm{f}}$ is the binding free energy of cluster (m) to cluster (n) and F_n^{f} is the formation free energy of cluster (n). This free energy notably contains a configurational entropy term. It is important to note that the thermodynamic part of a cluster dynamics model is only given by the emission coefficients. In particular, contrary to the classical nucleation theory (Feder et al. 1966), the chemical potential of the monomer in the matrix is not present. This is due to the fact that in cluster dynamics there is no distinction between the matrix, where heterophase fluctuations occur, and the secondary phase. All clusters, whatever their size, evolve due to the absorption and emission processes. This seamless description of the nucleation, growth, and coarsening processes is one of the strengths of the method.

If the hypothesis of infinite dilution is not well justified, corrections can be performed on the absorption (Nichols 1978; Brailsford and Bullough 1981) and on the emission (Lépinoux 2009; Berthier et al. 2010) rates.

Under irradiation, dislocation loops quickly reach sizes which are comparable to the mean distance between clusters. Loops interact with each other or with pre-existing dislocations. Loop size and density saturate, while the network dislocation density reaches a steady state, as a result of incorporation of loops and dislocation recovery (Brager et al. 1977). Cluster dynamics, as described above, is unable to simulate such phenomena. Although the validity of cluster dynamics at high doses is questionable, it may be desirable to include loop interactions and dislocation recovery into cluster dynamics formalism to perform qualitative modeling in support of experiments. This has been done recently (Zouari et al. 2011; Michaut et al. 2017), based on evolution equations proposed for rate theory (Stoller and Odette 1987) and then adapted for cluster dynamics (Jourdan 2015).

Up to now, cluster dynamics has been described for a pure metal containing self-defects or a dilute binary alloy under thermal aging. For more complex materials, a natural extension of Eq. (1) is to describe clusters through tuples $(n_1, n_2, \ldots, n_p)$, where p is the number of components in clusters. This has been done mostly in the case of self-defect clusters in the presence of gas atoms, such as helium and hydrogen (Ghoniem et al. 1983; Golubov et al. 2007; Ortiz et al. 2007; Marian and Bulatov 2011; Moll et al. 2013). For multicomponent alloys, one of the difficulties is to parametrize the model. In addition, negative flux couplings can arise, which cannot be taken into account in cluster dynamics due to its mean field character. Even if a reliable parametrization is available, the number of equations to solve becomes too large as soon as three components or more are present. For all these reasons, approximations to the initial problem can be necessary. Some recent attempts have aimed at simulating either precipitation or self-defect cluster evolution in multicomponent alloys (Ke et al. 2017; Mamivand et al. 2017), using equations similar to Eq. (1) coupled with CALPHAD approach.

Although surfaces can be taken into account in an effective way through sink strengths, it is sometimes necessary to describe them explicitly. For this purpose, a spatial dependency of rate equations on depth is introduced. The system is however assumed to be locally homogeneous, so that the mean field hypothesis of cluster dynamics remains valid. In practice, a diffusion term is added to Eq. (1) (Ortiz et al. 2004; Xu et al. 2012a; Jourdan et al. 2014), and the system is discretized along the depth. This approach is well adapted to simulate, for example, nonuniform damage creation by ion irradiation or irradiation of thin foils. Spatially resolved cluster dynamics in three dimensions has also been developed (Dunn and Capolungo 2015).

3 Parametrization of Cluster Dynamics

Among the parameters of cluster dynamics models, diffusion coefficients and binding energies at low temperature for small clusters can be estimated by atomistic calculations (Domain and Becquart 2001; Fu et al. 2004; Fu and Willaime 2008). Analytical laws for binding energies can be used to extrapolate values for larger clusters (Soneda and Diaz de la Rubia 2001; Varvenne et al. 2014; Alexander et al. 2016). A more difficult task is to calculate the binding *free* energies, so very often entropic contributions are neglected. However, it should be noted that to lead to quantitative results, such contributions should be included. In particular, it has been shown that cluster dynamics can reproduce atomistic kinetic Monte Carlo (AKMC) results on precipitation under thermal aging, provided that cluster free energies, taking into account configurational entropies, are calculated with the same energy model (Clouet et al. 2005; Jourdan et al. 2010) (Fig. 1). Another example is the annealing of loops in iron after helium implantation (Moll et al. 2013). Using an atomistically-informed model for the thermodynamics of helium bubbles (Jourdan and Crocombette 2011) appeared crucial to properly model the annealing kinetics of interstitial dislocation loops, which evolve due to Ostwald ripening by vacancy emission. Modeling the kinetics permitted, in turn, to *deduce* the vacancy free

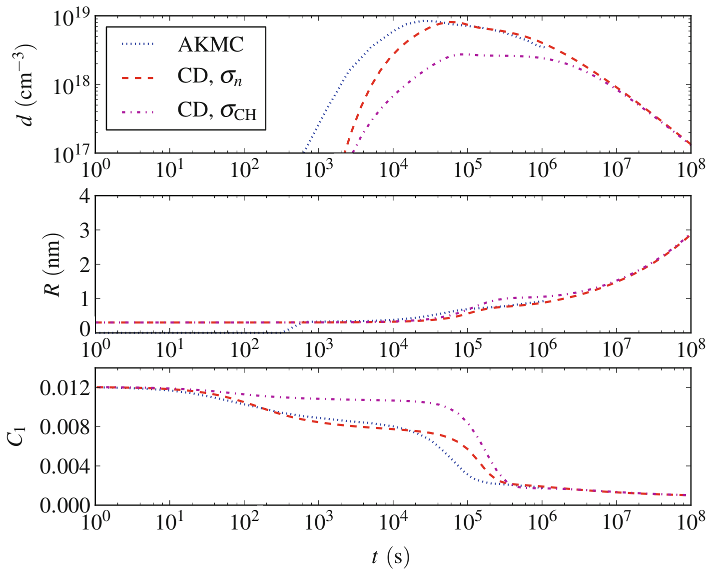

Fig. 1 Density of copper precipitates, mean radius, and monomer concentration in AKMC and cluster dynamics (CD) in a Fe-1.34at.%Cu alloy during thermal aging at 500 °C. Two different models for interface free energies in cluster dynamics are used, a Cahn-Hilliard approach (σ_{CH}) or a size-dependent interface energy (σ_n), obtained by the method of overlapping distributions, using the same energy model as for AKMC simulations (Jourdan et al. 2010)

activation energy from experiments. In general, under thermal aging, results are more sensitive to the precise values of emission coefficients than to the absorption coefficients.

For complex alloys, setting up a physically based thermodynamic parametrization is much more difficult than for model materials, if not impossible. A common approach is to use the so-called gray alloy approach. Only self-defect clusters are considered, and their properties reflect the chemical composition of the materials. Such an approach must somehow be calibrated on experiments and has a limited predictive character. However, it can be useful to perform sensitivity studies or to investigate the relevance of some mechanisms (Zouari et al. 2011; Michaut et al. 2017).

Under irradiation, microstructures of self-defect clusters are also very sensitive to coefficients $\beta_{n,m}$. Indeed, the growth rate of these clusters depends on the net flux of interstitials and vacancy to them. Since in general interstitials and vacancies are created in equal numbers, the growth of self-defect clusters results from tiny differences between interstitial and vacancy fluxes. These differences, which are crucial to explain phenomena such as void swelling and irradiation creep, are essentially explained by elastic effects which favor the absorption of interstitials by dislocations. In this “dislocation bias” model (Greenwood et al. 1959; Bullough and Perrin 1970), the preferential absorption of interstitials by dislocations leads, by conservation of matter, to a net flux of vacancies to voids. Therefore simulating quantitatively the growth of voids and dislocation loops requires precise knowledge of absorption coefficients. Obtaining sink strengths from experiments requires some assumptions (Kiritani et al. 1975; Golubov et al. 2001b), such as the fraction of freely migrating species. Alternatively, sink strengths can be evaluated numerically (Dubinko et al. 2005; Rouchette et al. 2014; Heinisch et al. 2000). It has been shown, however, that sink strengths greatly depend on the precise properties of point defects, such as point defect anisotropy at saddle point (Dederichs and Schroeder 1978; Skinner and Woo 1984; Carpentier et al. 2017), so up to now general laws to be implemented in cluster dynamics are still lacking.

Besides bias values, cluster dynamics results under irradiation are known to be very sensitive to production rates G_n. To define them, it is tempting to perform molecular dynamics (MD) simulations of displacement cascades, count the remaining clusters at the end of the simulations, and average the cluster populations over several runs. However, this method overestimates cluster densities in general. Indeed, at the end of MD simulations, clusters are still spatially correlated, whereas cluster dynamics can only describe a homogenous distribution of clusters. To circumvent this problem, an idea is to let self-defects and self-defect clusters diffuse over longer times, with OKMC, for example (Adjanor et al. 2010; Becquart and Domain 2011). It can be expected that if the temperature is high enough, the damage will be less correlated at the end of the simulation, so a transfer to cluster dynamics is more valid. However, it is not always easy to define a proper annealing time, since fast diffusers (typically interstitials) will homogenize more rapidly than slow diffusers (typically vacancies). For this reason it seems more legitimate to work on homogenization distance instead of homogenization time. This homogenization distance is given by cluster dynamics itself: for example, two clusters a few nanometers apart can be considered either as spatially correlated if the

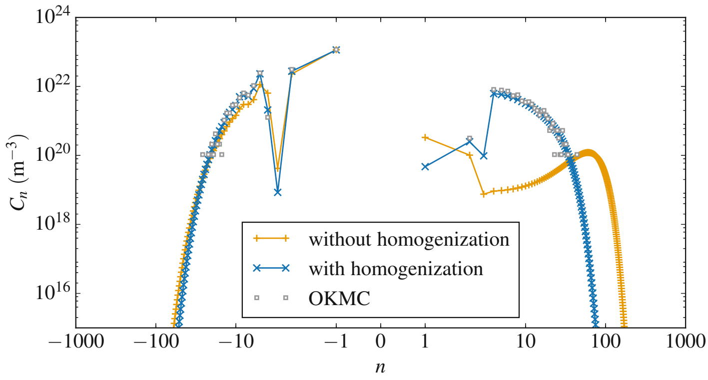

Fig. 2 Cluster distributions obtained with full OKMC simulations or cluster dynamics with two different source terms. Cascades are produced with the primary damage code MARLOWE (Robinson 1989), so only mono-interstitials and mono-vacancies are generated. For the non-homogenization case, a zero-time homogenization process was actually imposed to trigger close pair recombinations and clustering, but no thermally activated diffusion. Such small times are typical of MD simulations. Homogenization was performed using the procedure published in Jourdan and Crocombette (2012). Calculations were performed at 300 K up to 10^{-3} dpa, for an irradiation of iron with 20 keV iron ions and a damage rate of 10^{-4} dpa.s^{-1} (see Jourdan and Crocombette 2012 for more details)

density of clusters in cluster dynamics is low (e.g., at the beginning of irradiation) or uncorrelated if the average distance between defects in cluster dynamics is also a few nanometers. On this basis, a procedure has been proposed to provide a homogenized source term which is time dependent, even under constant irradiation (Jourdan and Crocombette 2012). A comparison between cluster dynamics with homogenized source term and full OKMC simulations, for an irradiation with 20 keV iron ions, is shown in Fig. 2. It should be noted that if only isolated Frenkel pairs are created by irradiation, this method is equivalent to reducing the production rate of self-interstitials and vacancies, due to recombinations which occur during homogenization. For ion and neutron irradiations, the source terms also contain self-defect clusters, either because they are created during the MD simulations or because they result from the diffusion and agglomeration of defects during the homogenization process. Therefore, this procedure naturally includes some aspects of "production bias" rate theory model (Golubov et al. 2012). One-dimensional migration of interstitial clusters, which is another cornerstone of this model, can be implemented by appropriate absorption coefficients (Singh et al. 1997).

4 Solving Cluster Dynamics Equations

Cluster concentrations at a given time are obtained by solving an ODE set containing equations similar to Eq. (1). This ODE set is known to be very stiff, which means that the use of explicit schemes requires very small time steps, so it may not be possible to run the problem until the desired physical time. It was soon recognized that implicit methods based on multistep methods, such as the GEAR package (Hindmarsh and Gelinas 1971) or the more recent CVODES solver (Hindmarsh et al. 2005), are more appropriate for such problems (Ghoniem and Sharafat 1980). These methods require the solving of linear systems based on the Jacobian matrix of the ODE set. Without optimization of this step, the calculation may be intractable for memory and computation time problems. For systems containing more than a few thousands of equations, it is clearly advantageous to provide the analytical form of the Jacobian matrix, instead of letting the solver evaluate it by finite differences. In addition, if only a small proportion of cluster classes are mobile, the Jacobian matrix is sparse, and dedicated solvers can be used. Examples of the use of sparse direct solvers are given in Xu et al. (2012a) and Jourdan et al. (2014). A final improvement consists in resizing the cluster size space on the fly, in order to avoid useless consideration of zero concentration clusters (Jourdan et al. 2014).

Even when the sparsity of the Jacobian matrix is taken into account, the computation time and memory use can become too large as the maximum cluster size increases. For example, to simulate the formation of a 10 nm cavity in iron filled with helium atoms over a spatial discretization of 100 slices, it is necessary to consider more than 10^{11} equations, which is well beyond the current capabilities of computation resources. In this case, approximating the master equation (1) becomes necessary. Two different kinds of methods can be distinguished:

- Grouping methods, first developed by Kiritani (1973), consist in solving equations for the moments of the cluster distribution over groups of equations. It was shown by Golubov et al. (2001a) and Ovcharenko et al. (2003) that accurate results can be obtained by taking into account the zeroth and first moments of the distribution. One of the advantages of this method is to be able to handle the mobility of large clusters (Golubov et al. 2007). However, it does not guarantee that cluster concentrations remain positive.
- Fokker-Planck-based methods. Although they can also be viewed as grouping methods since the final number of equations is reduced, the mathematical justification is completely different. For large values of n, the master equation for immobile species can be written as a Fokker-Planck equation (Goodrich 1964; Wolfer et al. 1977):

$$\frac{\partial}{\partial t}C(x,t) = -\frac{\partial}{\partial x}\left[F(x,t)C(x,t)\right] + \frac{\partial^2}{\partial x^2}\left[D(x,t)C(x,t)\right], \tag{5}$$

with $x_n = n$, $C(x_n, t) \approx C_n(t)$ and

$$F(x,t) = \sum_{m\in\mathscr{M}} m\left[\beta_m(x)C_m(t) - \alpha_m(x)\right], \tag{6}$$

$$D(x,t) = \frac{1}{2}\sum_{m\in\mathscr{M}} m^2\left[\beta_m(x)C_m(t) + \alpha_m(x)\right]. \tag{7}$$

This equation can then be discretized on a coarse mesh to reduce the number of equations to solve. Special care must be taken to connect the master equation used for small clusters to the discretized Fokker-Planck equation used for large clusters (Jourdan et al. 2014). Formulating the problem with a conservative form and using a finite volume approach is particularly convenient to ensure that matter is exactly conserved (Jourdan et al. 2014). The last important point is to choose an appropriate discretization scheme for the advection term. Recently, it has been shown that high-resolution, monotonicity-preserving schemes permit to accurately reproduce reference cluster distributions obtained by solving all equations, without introducing negative concentrations nor spurious oscillations (Jourdan et al. 2016) (Fig. 3). Although they are more computationally expensive than simple upwind schemes, high-resolution schemes do not introduce numerical diffusion and should therefore be preferred when high precision is needed on cluster

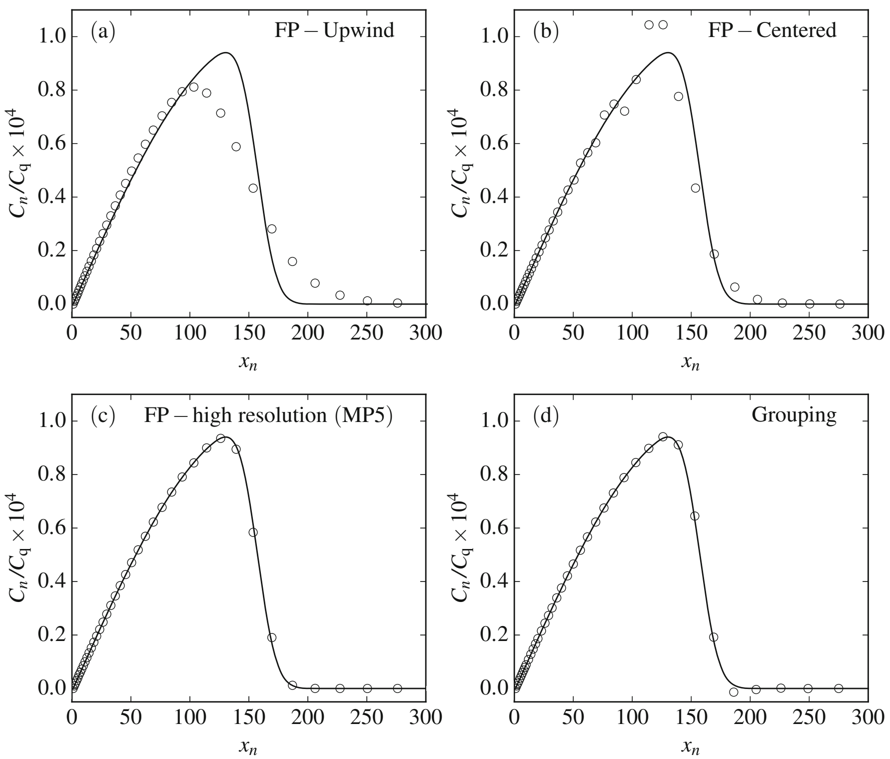

Fig. 3 Long-time limit cluster distributions for the clustering of quenched-in vacancies (Koiwa 1974) using different approximations (circles): (**a**, **b**, **c**) Fokker-Planck (FP) with upwind, centered, and high-resolution schemes for the advection term and (**d**) grouping method. Grouping and Fokker-Planck with high-resolution discretization scheme give very similar results, except in the distribution tail where negative concentrations occur with the grouping method (Jourdan et al. 2016)

distributions. The Fokker-Planck method can be generalized to multidimensional cases and permits to drastically reduce the number of equations. An example is given in Moll et al. (2013), where a spatialized simulation of helium implantation in iron could be performed. One drawback of this method, however, is that it is not really appropriate to treat the mobility of large clusters.

Alternatively, stochastic methods can be used to solve cluster dynamics equations. Among them, the simplest one is the so-called stochastic simulation algorithm (SSA) proposed by Gillespie (1976) and recently used for cluster dynamics (Marian and Bulatov 2011; Hoang et al. 2015). A finite volume V is envisaged, and equations on the number of clusters $N_n = C_n V$, derived from Eq. (1), are solved by a kinetic Monte Carlo method, using the appropriate rates for all possible events. An advantage of this method is that it does not require the solving of linear systems, so the memory requirements are lower than for deterministic solving, especially if large cluster spaces are considered. However, time steps are in general very low due to high-frequency events, so it is less efficient than deterministic solving for systems with one- or two-dimensional cluster spaces. Coupling stochastic approaches for large cluster sizes to deterministic solving for small clusters has been proposed, in order to avoid the stochastic treatment of high-frequency events (Surh et al. 2004; Gherardi et al. 2012; Terrier et al. 2017).

5 Conclusions and Perspectives

Over the last two decades, cluster dynamics has proven an interesting method to simulate irradiation processes over long time scales. Two aspects must be considered with care to obtain reliable results within the domain of validity of cluster dynamics: the numerical schemes to solve cluster dynamics equations and the parametrization of models. Efficient and accurate numerical schemes exist for time integration and for grouping methods. Additional efficiency improvements could be obtained by using hybrid methods, which couple deterministic to stochastic solving. Concerning the parametrization of cluster dynamics models, atomistic calculations now provide a large amount of data that can be used either directly in cluster dynamics or to parametrize scaling laws. Although no complete parametrization, from cluster free energies to sink strengths, exists even in pure metals, all tools are available to fill this gap. A remaining issue is to be able to treat heterogeneities in materials, which is not possible with current cluster dynamics models.

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